Relations

Staff

Column	Type	Key	Nullable
Staff_ID	INT	PK	No
First_Name	VARCHAR(50)		No
Last_Name	VARCHAR(50)		No
Email	VARCHAR(50)		No

Members

Column	Type	Key	Nullable
Member_ID	INT	PK	No
First_Name	VARCHAR(50)		No
Last_Name	VARCHAR(50)		No
Zip	CHAR(10)		No
Join_Date	DATE		No
Status	ENUM(‘active’, ‘suspended’)		No
Member_Email	VARCHAR(100)		No

Loans

Column	Type	Key	Nullable
Loan_ID	INT	PK	No
Tool_ID	INT	FK->Tool(Tool_ID)	No

Member_ID	INT	FK->Member(Member_ID)	No
Staff_ID	INT	FK->Staff(Staff_ID)	No
Out_Date	DATE		No
Due_Date	DATE		No
Back_Date	DATE		Yes
Fee_Charged	DECIMAL(8,2)		No

Tool

Column	Type	Key	Nullable
Tool_ID	INT	PK	No
Tool_Name	VARCHAR(50)		No
Tool_Brand	VARCHAR(50)		No
Category_ID	INT	FK->Category(Category_ID)	No
Purchased_Date	DATE		No
Price	DECIMAL(8,2)		No
Condition	ENUM('good', 'fair', 'needs repair', 'retired')		No
Tool_Loan_Status	ENUM(‘in’, ‘out’)		No

Categories

Column	Type	Key	Nullable
Category_ID	INT	PK	No

Category_Name	VARCHAR(20)		No
Daily_Late_Fee	DECIMAL(8,2)		No
Max_Loan_Days	INT		No
Requires_Certification	ENUM('Y', 'N')		No

Certification

Column	Type	Key	Nullable
Member_ID	INT	PK1	No
Category_ID	INT	PK2	No
Date_Earned	DATE		No

Normalization

Your checkout relation (the one that records who borrowed which tool): list its non-trivial functional dependencies, give its candidate key, and explain in one or two sentences why it is in 3NF.

Q1. *Relation*: Loan(**Loan_ID, Tool_ID, Member_ID, Staff_ID, Out_Date, Due_Date**, Back_Date, **Fee_Charged**)

Functional Dependencies: In the original, "messy" ERD, Tool_Name is a transitive dependency of Tool_ID, and Member_Name + Member_Zip are transitive dependencies (assumedly) of Member_Email.

Candidate Key: Loan_ID

Why it is in 3F: Our updated, "clean" ERD removed Tool and Member information, simplifying that to simply connect to the respective entities by Tool_ID and Member_ID. This removes the transitive and partial dependencies, as nothing depends on the primary key or a different key within the attribute. 3NF is classified as having no transitive or partial dependencies, hence our updated ERD is in 3NF.

Your certification relation (the one that records which members are certified for which categories): list its non-trivial functional dependencies, give its candidate key, and explain in one or two sentences why it is in 3NF.

Q2. *Relation*: Certification(**Member_ID, Category_ID,** Date_Earned)

Functional Dependencies: None

Candidate Key: Member_ID+Category_ID

Why it is in 3F: None of the attributes within the entity are dependent on each other. There is no entity reliant on the primary key, meaning there is no partial dependency. There is also no entity reliant on a non-primary key, meaning there is no transitive dependency. The lack of dependency, both in the “messy” and “clean” ERDs means that it is in 3NF, as 3NF lacks dependencies.

The name-change problem. Your memo describes a member whose name had to be fixed in many rows of the old loan log (`loans.csv`). Which functional dependency caused that problem? Which normal form did the loan log violate because of it? How does your schema prevent the problem?

Q3. In the original design, the loan log records the *member_email*, *member_name*, and *member_zip*. This is a transitive dependency violating 3NF between those three values. The schema prevents the problem by only including *member_id* in the loan log, and keeping all other member attributes within the member entity. This ensures that the member_email, for example, must only change once in the member entity and all other instances of the member_email in the loan log will update from that singular change point, rather than having to update each instance in loans as previously established.